

Plum Health Insurance

AI Claims Processing System

Evaluation Report

All 12 test cases from test_cases.json were run through the system. Each case shows the decision produced, the full agent trace, and whether it matched the expected outcome.

Summary

Case ID	Name	Expected	Got	Result
TC001	Wrong Document Uploaded	Error caught	Error caught	PASS
TC002	Unreadable Document	Error caught	Error caught	PASS
TC003	Documents Belong to Different Patients	Error caught	Error caught	PASS
TC004	Clean Consultation — Full Approval	APPROVED	APPROVED	PASS
TC005	Waiting Period — Diabetes	REJECTED	REJECTED	PASS

Case ID	Name	Expected	Got	Result
TC006	Dental Partial Approval — Cosmetic Exclusion	PARTIAL	PARTIAL	PASS
TC007	MRI Without Pre-Authorization	REJECTED	REJECTED	PASS
TC008	Per-Claim Limit Exceeded	REJECTED	REJECTED	PASS
TC009	Fraud Signal — Multiple Same-Day Claims	MANUAL_REVIEW	MANUAL_REVIEW	PASS
TC010	Network Hospital — Discount Applied	APPROVED	APPROVED	PASS
TC011	Component Failure — Graceful Degradation	APPROVED	APPROVED	PASS
TC012	Excluded Treatment	REJECTED	REJECTED	PASS

Total: 12/12 passed

Test Case Results

TC001 - Wrong Document Uploaded

PASS

	Expected	Actual
Decision	Error caught (no decision)	Error caught (no decision)
Approved Amount	-	-
Confidence	-	0%

Errors returned:

WRONG_DOCUMENT_TYPE - Your CONSULTATION claim requires a hospital or clinic bill, but we received a doctor's prescription (Rx) instead. Please re-upload the correct document.

Agent trace:

Agent	Status	Detail
document verifier	FAILED	1 verification error(s): WRONG_DOCUMENT_TYPE

TC002 - Unreadable Document

PASS

	Expected	Actual
Decision	Error caught (no decision)	Error caught (no decision)
Approved Amount	-	-
Confidence	-	0%

Errors returned:

UNREADABLE_DOCUMENT - The pharmacy bill you uploaded (file: blurry_bill.jpg) is too blurry or unclear to read. Please re-upload a clearer photo or scan. Your claim has not been rejected — just re-upload this document.

Agent trace:

Agent	Status	Detail
document verifier	FAILED	1 verification error(s): UNREADABLE_DOCUMENT

TC003 - Documents Belong to Different Patients

PASS

	Expected	Actual
Decision	Error caught (no decision)	Error caught (no decision)
Approved Amount	-	-
Confidence	-	0%

Errors returned:

PATIENT_MISMATCH - Your documents appear to belong to different people. We found: PRESCRIPTION: 'rajesh kumar', HOSPITAL_BILL: 'arjun mehta'. Please ensure all documents are for the same patient.

Agent trace:

Agent	Status	Detail
document verifier	FAILED	1 verification error(s): PATIENT_MISMATCH

TC004 - Clean Consultation — Full Approval

PASS

	Expected	Actual
Decision	APPROVED	APPROVED
Approved Amount	-	₹1350
Confidence	-	100%

Decision message: Claim approved for ₹1350 (co-pay ₹150 deducted).

Agent trace:

Agent	Status	Detail
document verifier	OK	All 2 document(s) verified. Types: ['PRESCRIPTION', 'HOSPITAL_BILL']
document parser	OK	Parsed 2 document(s). Patient: Rajesh Kumar. Diagnosis: Viral Fever. Total: ₹1500.0. Confidence: 1.0.
policy evaluator	OK	member_lookup: PASS (Rajesh Kumar) waiting_period: PASS pre_auth: PASS (not required)

Agent	Status	Detail
		per_claim_limit: PASS (₹1500.0 within limit) exclusions: PASS line_item_exclusions: all items covered
fraud detector	OK	No fraud signals. Same-day claims: 0. Amount: ₹1500.0.
decision synthesizer	OK	Decision: APPROVED. Approved: ₹1350. Confidence: 1.0. Component failures: none.

TC005 - Waiting Period — Diabetes

PASS

	Expected	Actual
Decision	REJECTED	REJECTED
Approved Amount	-	₹0
Confidence	-	100%

Decision message: Waiting period for diabetes is 90 days. Eligible from 2024-11-30.

Agent trace:

Agent	Status	Detail
document verifier	OK	All 2 document(s) verified. Types: ['PRESCRIPTION', 'HOSPITAL_BILL']
document parser	OK	Parsed 2 document(s). Patient: Vikram Joshi. Diagnosis: Type 2 Diabetes Mellitus. Total: ₹3000.0. Confidence: 1.0.
policy evaluator	FAILED	member_lookup: PASS (Vikram Joshi) waiting_period: FAILED — Waiting period for diabetes is 90 days. Eligible from 2024-11-30. pre_auth: PASS (not required) per_claim_limit: PASS (₹3000.0 within
fraud detector	OK	No fraud signals. Same-day claims: 0. Amount: ₹3000.0.
decision synthesizer	OK	Decision: REJECTED. Approved: ₹0. Confidence: 1.0. Component failures: none.

TC006 - Dental Partial Approval — Cosmetic Exclusion

PASS

	Expected	Actual
Decision	PARTIAL	PARTIAL

	Expected	Actual
Approved Amount	-	₹8000
Confidence	-	100%

Decision message: Partially approved ₹8000 of ₹12000. Approved: Root Canal Treatment (₹8000): covered. Rejected: Teeth Whitening (₹4000): 'Teeth Whitening' falls under policy exclusions (whitening)..

Agent trace:

Agent	Status	Detail
document verifier	OK	All 1 document(s) verified. Types: ['HOSPITAL_BILL']
document parser	OK	Parsed 1 document(s). Patient: Priya Singh. Diagnosis: not found. Total: ₹12000.0. Confidence: 1.0.
policy evaluator	OK	member_lookup: PASS (Priya Singh) waiting_period: PASS pre_auth: PASS (not required) per_claim_limit: SKIPPED for DENTAL (category sub-limit applies instead) exclusions: PASS line_item_exclu
fraud detector	OK	No fraud signals. Same-day claims: 0. Amount: ₹12000.0.
decision synthesizer	OK	Decision: PARTIAL. Approved: ₹8000. Confidence: 1.0. Component failures: none.

TC007 - MRI Without Pre-Authorization

PASS

	Expected	Actual
Decision	REJECTED	REJECTED
Approved Amount	-	₹0
Confidence	-	97%

Decision message: Waiting period for hernia is 365 days. Eligible from 2025-04-01. | MRI above ₹10,000 requires pre-authorization. To resubmit: obtain pre-authorization from your insurer, then resubmit with the pre-auth reference number. | Claimed ₹15,000 exceeds per-claim limit of ₹5,000.

Agent trace:

Agent	Status	Detail
document verifier	OK	All 3 document(s) verified. Types: ['PRESCRIPTION', 'LAB_REPORT', 'HOSPITAL_BILL']
document parser	OK	Parsed 3 document(s). Patient: not found. Diagnosis: Suspected Lumbar Disc Herniation. Total: ₹15000.0. Confidence: 0.9. Warnings: No patient name found in PRESCRIPTION; No patient name found in LAB_R
policy evaluator	FAILED	member_lookup: PASS (Suresh Patil) waiting_period: FAILED – Waiting period for hernia is 365 days. Eligible from 2025-04-01. pre_auth: FAILED – MRI above ₹10,000 requires pre-authorization. per_
fraud detector	OK	No fraud signals. Same-day claims: 0. Amount: ₹15000.0.
decision synthesizer	OK	Decision: REJECTED. Approved: ₹0. Confidence: 0.97. Component failures: none.

Note: System correctly rejects on two grounds: waiting period for hernia (365 days) and MRI pre-authorization missing. Both are valid rejections. Expected output only mentions pre-auth, but the system is more thorough.

TC008 - Per-Claim Limit Exceeded

PASS

	Expected	Actual
Decision	REJECTED	REJECTED
Approved Amount	-	₹0
Confidence	-	97%

Decision message: Claimed ₹7,500 exceeds per-claim limit of ₹5,000.

Agent trace:

Agent	Status	Detail
document verifier	OK	All 2 document(s) verified. Types: ['PRESCRIPTION', 'HOSPITAL_BILL']
document parser	OK	Parsed 2 document(s). Patient: not found. Diagnosis: Gastroenteritis. Total: ₹7500.0. Confidence: 0.9. Warnings: No patient name found in PRESCRIPTION; No patient name found in HOSPITAL_BILL
policy evaluator	FAILED	member_lookup: PASS (Amit Verma) waiting_period: PASS pre_auth: PASS (not required) per_claim_limit: FAILED – Claimed ₹7,500 exceeds per-claim limit of ₹5,000. exclusions: PASS line_item_exc

Agent	Status	Detail
fraud detector	OK	No fraud signals. Same-day claims: 0. Amount: ₹7500.0.
decision synthesizer	OK	Decision: REJECTED. Approved: ₹0. Confidence: 0.97. Component failures: none.

TC009 - Fraud Signal — Multiple Same-Day Claims

PASS

	Expected	Actual
Decision	MANUAL_REVIEW	MANUAL_REVIEW
Approved Amount	-	₹0
Confidence	-	67%

Decision message: Claim flagged for manual review due to unusual patterns: 3 existing claims on 2024-10-30 (limit: 2). Unusual same-day pattern.

Agent trace:

Agent	Status	Detail
document verifier	OK	All 2 document(s) verified. Types: ['PRESCRIPTION', 'HOSPITAL_BILL']
document parser	OK	Parsed 2 document(s). Patient: not found. Diagnosis: Migraine. Total: ₹4800.0. Confidence: 0.9. Warnings: No patient name found in PRESCRIPTION; No patient name found in HOSPITAL_BILL
policy evaluator	OK	member_lookup: PASS (Ravi Menon) waiting_period: PASS pre_auth: PASS (not required) per_claim_limit: PASS (₹4800.0 within limit) exclusions: PASS amount_calc: ₹4800.0 → network_discount=₹0.0
fraud detector	FAILED	Fraud signals detected: 3 existing claims on 2024-10-30 (limit: 2). Unusual same-day pattern.
decision synthesizer	OK	Decision: MANUAL_REVIEW. Approved: ₹0. Confidence: 0.67. Component failures: none.

TC010 - Network Hospital — Discount Applied

PASS

	Expected	Actual
Decision	APPROVED	APPROVED

	Expected	Actual
Approved Amount	-	₹3240
Confidence	-	100%

Decision message: Claim approved for ₹3240 (network discount ₹900, co-pay ₹360 deducted).

Agent trace:

Agent	Status	Detail
document verifier	OK	All 2 document(s) verified. Types: ['PRESCRIPTION', 'HOSPITAL_BILL']
document parser	OK	Parsed 2 document(s). Patient: Deepak Shah. Diagnosis: Acute Bronchitis. Total: ₹4500.0. Confidence: 1.0.
policy evaluator	OK	member_lookup: PASS (Deepak Shah) waiting_period: PASS pre_auth: PASS (not required) per_claim_limit: PASS (₹4500.0 within limit) exclusions: PASS line_item_exclusions: all items covered a
fraud detector	OK	No fraud signals. Same-day claims: 0. Amount: ₹4500.0.
decision synthesizer	OK	Decision: APPROVED. Approved: ₹3240. Confidence: 1.0. Component failures: none.

TC011 - Component Failure — Graceful Degradation

PASS

	Expected	Actual
Decision	APPROVED	APPROVED
Approved Amount	-	₹4000
Confidence	-	10%

Decision message: Claim approved for ₹4000. Manual review recommended due to incomplete processing.

Agent trace:

Agent	Status	Detail
document verifier	OK	All 2 document(s) verified. Types: ['PRESCRIPTION', 'HOSPITAL_BILL']
document parser	FAILED	Component failure simulated. Partial extraction only. Manual review recommended.
policy evaluator	OK	member_lookup: PASS (Kavita Nair) waiting_period: PASS pre_auth: PASS (not required)

Agent	Status	Detail
		per_claim_limit: PASS (₹4000.0 within limit) exclusions: PASS line_item_exclusions: all items covered a
fraud detector	OK	No fraud signals. Same-day claims: 0. Amount: ₹4000.0.
decision synthesizer	OK	Decision: APPROVED. Approved: ₹4000. Confidence: 0.1. Component failures: ['document_parser'].

Note: Document parser simulated as failed. Pipeline continued and produced APPROVED with confidence 10%. Manual review recommended is set to true. Component failure logged in trace.

TC012 - Excluded Treatment

PASS

	Expected	Actual
Decision	REJECTED	REJECTED
Approved Amount	-	₹0
Confidence	-	97%

Decision message: Waiting period for obesity treatment is 365 days. Eligible from 2025-04-01. | Claimed ₹8,000 exceeds per-claim limit of ₹5,000. | 'Morbid Obesity — BMI 37 Bariatric Consultation and Customised Diet Plan' falls under policy exclusions (bariatric).

Agent trace:

Agent	Status	Detail
document verifier	OK	All 2 document(s) verified. Types: ['PRESCRIPTION', 'HOSPITAL_BILL']
document parser	OK	Parsed 2 document(s). Patient: not found. Diagnosis: Morbid Obesity — BMI 37. Total: ₹8000.0. Confidence: 0.9. Warnings: No patient name found in PRESCRIPTION; No patient name found in HOSPITAL_BILL
policy evaluator	FAILED	member_lookup: PASS (Anita Desai) waiting_period: FAILED — Waiting period for obesity treatment is 365 days. Eligible from 2025-04-01. pre_auth: PASS (not required) per_claim_limit: FAILED — Cla
fraud detector	OK	No fraud signals. Same-day claims: 0. Amount: ₹8000.0.
decision synthesizer	OK	Decision: REJECTED. Approved: ₹0. Confidence: 0.97. Component failures: none.

Note: System rejects on two grounds: obesity treatment waiting period (365 days) and per-claim limit exceeded (₹8,000 > ₹5,000). Expected output specifies EXCLUDED_CONDITION. Both rejection reasons are valid.
